LSTM Multi-Step Forecasting Results & Insights

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1 System Architecture & Setup

- **Input Window:** Fixed at 10 lookback timesteps (≈ 27.8 ms of history)
- **Target Output Horizons:** Evaluated for 10 steps (≈ 27.8 ms), 20 steps (≈ 55.6 ms), and 50 steps (≈ 138.9 ms).
- **Model Architecture:** Direct Multi-Output structure:

$$\text{Input}(10, 1) \longrightarrow \text{LSTM}(64) \longrightarrow \text{Dense}(32, \text{ReLU}) \longrightarrow \text{Dense}(H)$$

- **Dataset Split:** 21 patients (excluding outlier records 111 & 118). Per patient: 40,000 steps for training, 10,000 steps for validation, and 50,000 steps for testing.

2 Experimental Results

The model was trained independently per patient with early stopping (patience = 5). Metrics are averaged across all 21 patients(intra-patient).

Table 1: LSTM Multi-Step Forecasting Results Across Prediction Horizons

Horizon (Steps)	Time Window	RMSE	MAE	R^2 Score	Comp. Time (s)	Comp. Time (
10	≈ 27.8 ms	**0.0418**	**0.0205**	**0.8241**	687.7 s	≈ 11.5 min
20	≈ 55.6 ms	**0.0637**	**0.0310**	**0.6160**	747.4 s	≈ 12.5 min
50	≈ 138.9 ms	**0.0842**	**0.0478**	**0.3596**	767.5 s	≈ 12.8 min

3 Visualizations

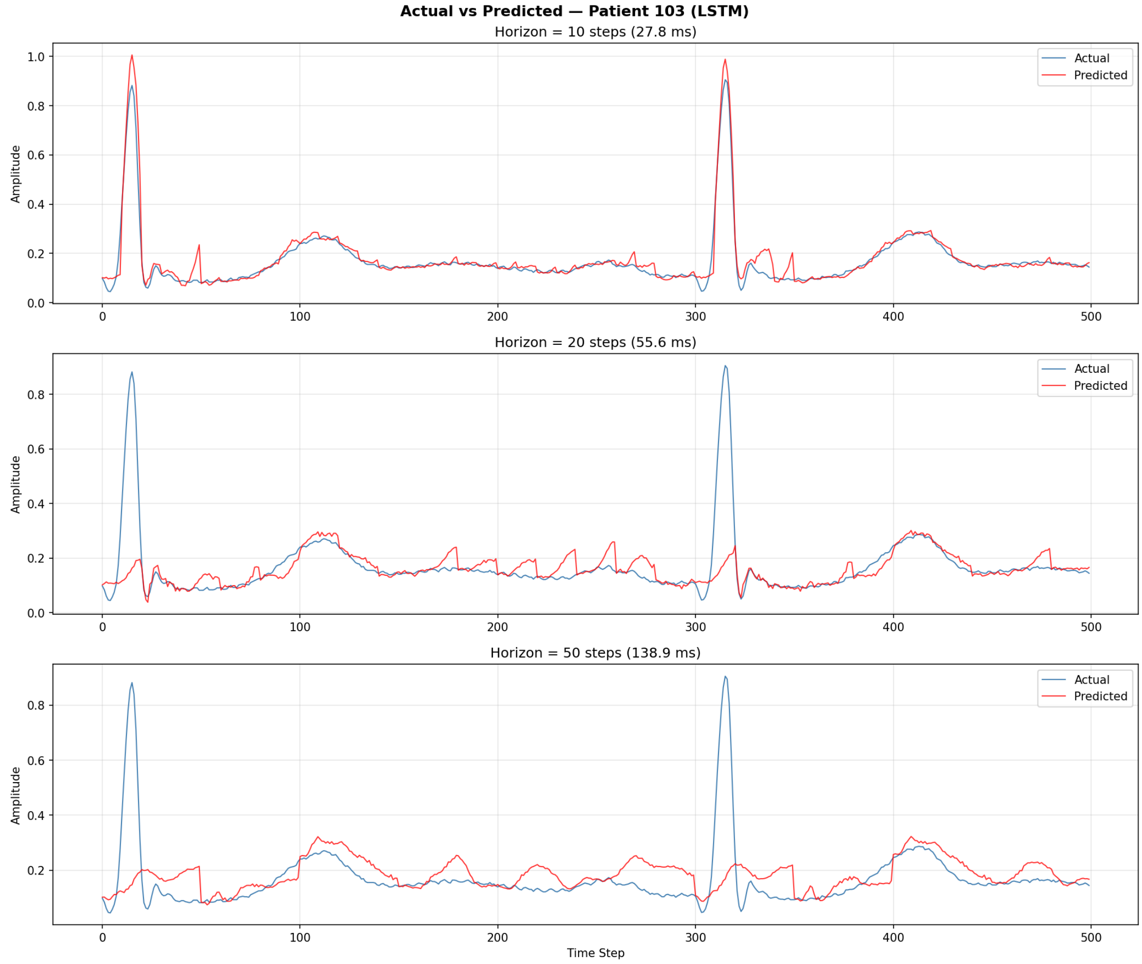

Figure 1: Actual vs. Predicted ECG Signals for Patient 103 across Horizons 10, 20, and 50.

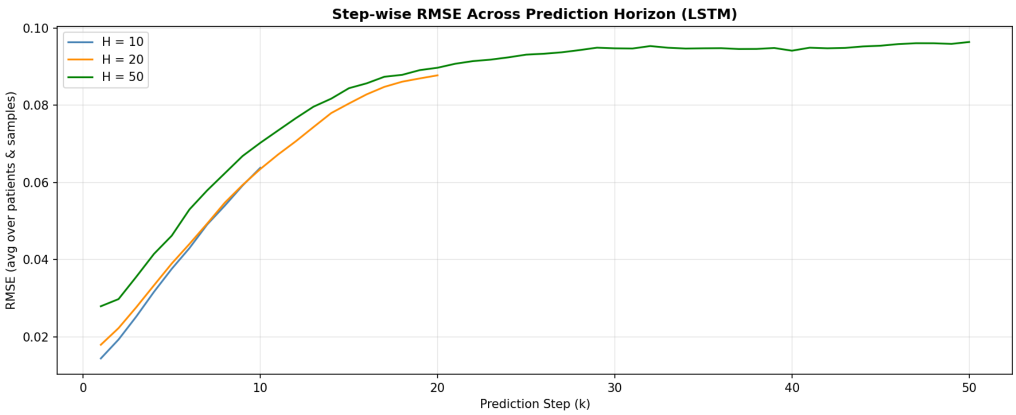

Figure 2: Step-wise RMSE growth across prediction steps $k = 1 \dots H$.

4 Key Insights & Proposed Research Direction

4.1 Error Accumulation & The MSE Flaw

As seen in Figure 1 and Figure 2, performance degrades significantly as the prediction horizon expands from 10 to 50 steps (R^2 drops from 0.8241 to 0.3596).

The root cause lies in the interaction between **long-horizon uncertainty** and standard **Mean Squared Error (MSE)** loss:

1. MSE penalizes large errors quadratically (e^2).
2. When predicting 50 steps into the future from only 10 steps of history, uncertainty increases rapidly.
3. To minimize expected MSE loss under high uncertainty, the model learns a conservative strategy: it predicts the mathematical mean amplitude.
4. As a result, sharp ECG peaks (such as the QRS complex) are smoothed out and flattened.

4.2 Clinical Significance

From a medical/cardiologist perspective, an ECG prediction that smooths out or misses QRS complexes is **clinically useless**. Diagnostic indicators (such as R-peak amplitude, ST-segment elevation, and arrhythmia detection) rely entirely on sharp morphological features.

4.3 Proposed Research Direction

Standard Euclidean losses (MSE / MAE) fail for long-horizon ECG forecasting because they ignore temporal alignment and shape dynamics. My proposed research direction focuses on:

- **DILATE Loss (DIStortion Loss with Adaptive Time Warp Estimation):** Combining Dynamic Time Warping (DTW) shape loss with temporal penalty loss to preserve peak heights and alignment. This is my proposed research direction that I am actively working on right now .